Constructions of Covering Sequences and 2D-sequences

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Abstract

An (n, R)-covering sequence is a cyclic sequence whose consecutive n-tuples form a code of length n and covering radius R. Using several construction methods improvements of the upper bounds on the length of such sequences for $n \leq 20$ and $1 \leq R \leq 3$, are obtained. The definition is generalized in two directions. An (n, m, R)-covering sequence code is a set of cyclic sequences of length m whose consecutive n-tuples form a code of length n and covering radius R. The definition is also generalized to arrays in which the $m \times n$ sub-matrices form a covering code with covering radius R. We prove that asymptotically there are covering sequences that attain the sphere-covering bound up to a constant factor.

I. Introduction

An (n, R)***-covering code*** $\mathcal{C}$ is a set of words of length n over a given alphabet Σ_q, of size q, such that each word of length n over Σ_q is within distance R from at least one codeword in $\mathcal{C}$. In other words, for each $x \in \Sigma_q^n$, there exists $c \in \mathcal{C}$ such that $d(x, c) \leq R$, where $d(y, z)$, $y, z \in \Sigma_q^n$, denotes the Hamming distance between y and z. Covering code were always of interest, but the interest increased due to the following three seminal papers [12], [13], [31]. The interest was also increased partially because of the connection of covering codes to data compression. An excellent book that covers all aspects of such codes is [11].

Chung and Cooper [10] generalized the notion of an (n, R)-covering code of length n and radius R to a cyclic sequence whose consecutive n-tuples form a covering code of length n and radius R. They called such a structure a de Bruijn covering code since n-tuples of a cyclic sequence are considered and this sequence forms a cycle in the de Bruijn graph. However, any cyclic sequence can be viewed as a cycle in the de Bruijn graph, and the sequence itself has a very loose connection to the de Bruijn graph, although some of the constructions in the paper use ingredients and concepts from the de Bruijn graph. Therefore, we prefer to call such a sequence a ***covering sequence***. This is a dual definition of what is known as a robust sequence, i.e., a sequence whose consecutive n-tuples form an error-correcting code of length n and minimum distance d. Such sequences as well as arrays have been considered for example in [1], [7], [8], [32], [38], [41], [52], [55] and they have variety of applications.

All the discussion from Section III will be restricted to the binary alphabet and hence alphabet size will rarely be mentioned. However, many of the ideas that will be presented can be generalized quite easily to any alphabet size. An (n, R)***-covering sequence*** (an (n, R)-CS for short) is a cyclic sequence whose consecutive n-tuples form an (n, R)-covering code. This structure was first defined and analyzed in [10]. To find the shortest (n, R)-CS, denoted by $\mathcal{L}(n, R)$, we define two more related structures of cyclic sequence codes. A cyclic covering sequence code contains sequences in which the consecutive n-tuples of all the sequences form an (n, R)-covering code. Such codes will be discussed in our exposition.

In recent years many one-dimensional coding problems have been considered in the two-dimensional framework due to modern applications. This is quite natural and has become fashionable from both theoretical and practical points of view. Such generalizations were considered for various structures such as error-correcting codes [2], [45], burst-correcting codes [3], [5], [26], [27], constrained codes [46],

[51], de Bruijn sequences [18], [21], [40], [43], M-sequences where the two dimensional sequences are pseudo-random arrays [20], [39]. Other structures designed for applications are robust self-location arrays with window property [7] and structured-light patterns [42]. Therefore, it is very tempting to generalize the concept of covering sequences into a two-dimensional framework and this is one of the targets of the current work.

An $(m \times n, R)$***-covering 2D-sequence*** (an $(m \times n, R)$-C2DS for short) is a doubly-periodic $M \times N$ array (cyclic horizontally and vertically like a torus) over an alphabet of size q such that the set of all its $m \times n$ windows form a covering code with radius R.

While in the one-dimensional case we are interested in the (n, R)-CS of the shortest length, in the two-dimensional case we are interested in the $(m \times n, R)$-C2DS with the smallest area, but the ratio between M and N can be important too.

Remark 1. It is tempting to use the term "covering array" for the two-dimensional matrices, but this term is already reserved to another combinatorial object associated with covering, see [14], [15], [47] and references therein.

Our goals in this paper are to present construction methods for covering sequences, covering sequence codes, and covering 2D-sequences. Some of the constructions yield sequences and codes that are almost optimal.

The rest of the paper is organized as follows. In Section II, some basic results and important known results on covering codes and covering sequences are presented. Section III is devoted to cyclic covering sequence codes and the constructions of covering sequences from cyclic covering sequence codes. In Section IV a cyclic covering sequence code based on self-dual sequences is presented. Based on this code a relatively short $(2^k, 1)$-CS whose length is within factor of 1.25 from the sphere-covering bound is obtained. An interleaving construction to obtain an (n, R)-CS from an (n_1, R_1)-CS and an (n_2, R_2)-CS, where $n = n_1 + n_2$ and $R = R_1 + R_2$ is presented in Section V. When $n_1 = n_2$ and $R_1 = R_2$ a better interleaving construction is presented. A construction based on primitive polynomials, with a certain structure, is presented in Section VI. Section VII is the only one devoted to covering 2D-sequences and introduces two methods to generate such arrays: one is by folding a covering sequence and the other is by making all possible shifts of a related covering sequence. These methods are used to obtain upper bounds on the size of such arrays and to construct them. Finally, conclusion and further problems for future research are presented in Section VIII.

II. Preliminaries

The area of covering codes is very well established in information theory as a dual for error-correcting codes. Bounds on the sizes of such codes were extensively studied, where the upper bounds are either by constructions or using probabilistic methods to prove the existence of some codes asymptotically. Lower bounds are usually obtained by analytic methods, but they are usually not much better than the sphere-covering bound which is the most basic bound.

A ball of radius R around a word $\boldsymbol{x}$ of length n is the set of words whose distance is at most R from $\boldsymbol{x}$. The size of such ball denoted by $V_q(n, R)$ is

$$V_q(n, R) = \sum_{i=0}^{R} \binom{n}{i} (q-1)^i \ .$$

Since the balls of radius R around the codewords of an (n, R)-covering code $\mathcal{C}$ contain the whole space, it follows that a lower bound on the size of $\mathcal{C}$ is

$$|\mathcal{C}| \geq \frac{q^n}{V_q(n, R)}.$$

This bound is the ***sphere-covering bound***. There exists a covering code that approaches this bound up to a factor roughly $eR\log R$ [37]. The proof method for this bound is probabilistic. A similar bound for an (n,R)-CS over a prime power alphabet was presented in [10]. This bound was generalized to any alphabet by Vu [54]. The bound states that for fixed R there exists an (n,R)-CS, over Σ_q, whose length is at most $\mathcal{O}\left(\frac{q^n}{V_q(n,R)}\log n\right)$.

This result of Vu [54] was generalized in our conference paper [9] where we proved the following

Proposition 2. *Let m,n be nonnegative integers. For any $M \geq m$, there exists an $M\times N$ $(m\times n,R)$-C2DS such that $M\cdot N=\mathcal{O}\left(\frac{q^{mn}}{V_q(mn,R)}\cdot(\log m+\log n)\right)$, for fixed q and R.*

The proof of Proposition 2 was presented in [9] using carefully the probabilistic method which was also used in [54] and will not be proved here. The same existence proof can be obtained by applying folding technique which will be presented in Section VII on the one-dimensional sequences which are known to exist by the asymptotic bound of [54].

For small R, there are some (n,R)-covering codes that attain the sphere-covering bound with equality, such as the Hamming codes of length $n=2^k-1$ and radius one. Other codes are very close to the upper bound, such as the ones for $R=2$ which are perfect asymptotically [50, Construction 4.24] or other similar codes [22], [24]. Similarly, such sparse covering codes were also considered for $R=3$ [22], [24]. The main goal of the research on (n,R)-CSs is to get as close as possible to the upper bounds obtained for covering codes.

One of our constructions will use a span n de Bruijn sequence over Σ_q. This is a cyclic sequence of length q^n, where each n-tuple over Σ_q is contained in exactly one window of consecutive digits in the sequence. Such sequences exist for all $q\geq 2$ and $n\geq 1$.

Finally, one very trivial bound which was mentioned in [10] will be used in one of the best known bounds (see Table I).

Theorem 3. *For any $n,R\geq 1$ we have that $\mathcal{L}(n,R)\leq\mathcal{L}(n+1,R)$.*

III. Construction from a Cyclic Covering Code

From this section, the discussion will be focused on binary alphabet, but some of the constructions can be adapted to nonbinary alphabet. This is especially true for this section where all the results can be given also for codes over an arbitrary alphabets. Cyclic covering codes are perhaps the most important ingredients in constructions of (n,R)-CSs, especially when n and R are relatively small. They were considered in various papers, e.g. [16], [17], [33], [36]. There are two definitions for covering sequence codes.

An (n,m,R)***-CS code*** (an (n,m,R)-CSC for short) $\mathcal{C}$ is a set of cyclic codewords of length m such that each word of length n is within distance R from at least one n-tuple of a codeword of $\mathcal{C}$, i.e., the consecutive n-tuples of all the codewords form an (n,R)-covering code.

An (n,R)***-CS code*** (an (n,R)-CSC for short) $\mathcal{C}$ is a set of cyclic words of possibly different lengths such that each word of length n is within distance R from at least one window of length n in one of the sequences, i.e., the consecutive n-tuples of all the codewords form an (n,R)-covering code.

The only distinction between the two types of codes is that in the first one of an (n,m,R)-CSC, all codewords have the same length m and in the second one of an (n,R)-CSC codewords can be of different lengths. The (n,R)-CS is a cyclic sequence and hence it is a cyclic covering sequence code with one codeword.

The natural criteria to compare different (n,R)-CSCs is the total length of all the sequences in each code. Another criteria for comparison between CSCs which is important in our exposition is the length of the (n,R)-CS that can be constructed by combining the codewords of each CSC. This process is done as follows.

The sequences of the code are combined by first concatenating the first $n-1$ bits of each sequence after its last bit. This converts a cyclic sequence to an acyclic sequence, where both sequences have the same n-tuples. After the sequences become acyclic, we order the sequences in a cyclic list, where two consecutive sequences in the list overlap such that the suffix of the first sequence aligns with the prefix of the next sequence. The target is to have the total overlaps (sum of the lengths of all the overlaps) as large as possible.

For this purpose, when we concatenate the suffix of length $n-1$ after the last bit of the codeword, we should try to do it for every cyclic shift of the codeword and ultimately use the shifts that yield the largest total overlaps. Merging all the sequences together is done following the order of the list, where each overlap is taken naturally only once. Finally, there might be sequences in the code with periodicity. The periodicity should be eliminated either within the code or in the final (n, R)-CS. This is demonstrated in the following example.

Example 4. The following eight codewords form a $(9, 10, 1)$-CSC $\mathcal{C}$:

$$[1000010000], [0001001101], [1001111001], [1111010111],$$
$$[1010101010], [0101011000], [0110111001], [0111010000].$$

We extend each cyclic sequence by eight bits to obtain all the 9-tuples of the cyclic sequences in acyclic sequences. We furthermore find the order of these eight sequences in a way that the total overlap between suffixes and prefixes of the consecutive sequences (including the last sequence in the list and the first sequence in the list) will be large as possible, where in this process all possible shifts of all the sequences are considered. The obtained eight acyclic sequences of length 18 are as follows, where the overlap is written after the sequence:

the sequence	*overlap*
100001000010000100	6
000100110100010011	5
100111100110011110	5
111101011111110101	5
101010101010101010	5
010101100001010110	4
011011100101101110	5
011101000001110100	3

The total number of bits in the eight sequences is 144 and the total number of overlaps is 38. This yield a $(9, 1)$-CS of length $144 - 38 = 106$ as follows:

$$[100001000010000100110100010011110011001111010111111110$$
$$10101010101010101100001010110111001011011101000001110]$$

However, the codeword $[1000010000]$ of $\mathcal{C}$ is periodic and contains only five 9-tuples as the cyclic sequence $[10000]$. Moreover, the codeword $[1010101010]$ of $\mathcal{C}$ is also periodic and it contains only two 9-tuples as the cyclic sequence $[10]$. Hence, they are redundant in the code and in the final $(9, 1)$-CS. Therefore, we can eliminate the redundancy in the covering sequence and obtain the following $(9, 1)$-CS of length $106 - 13 = 93$:

$$[100001000010011010001001111001100111101011111110$$
$$10101010110000101011011100101101110100000 1110]$$

Introducing the codewords $[10000]$ instead of $[1000010000]$ and $[10]$ instead of $[1010101010]$ in the code $\mathcal{C}$ yields a $(9,1)$-CSC with 6 codewords of length 10, one codeword of length 5, and one codeword of length 2.

This sequence of length 93 can be extended to a $(9,1)$-CS of length 102 with 8 consecutive *ones* which will be required later:

$$[100001000010011010001001111001100111101011111111011111110$$
$$10101010110000101011011100101101110100000111 0]$$

■

Many of the shortest (n,R)-CSs for small n and R were found by computer search. For this purpose we use the well-known problem of the ***Shortest Cyclic Superstring (SCS)***. This problem is designed to find the shortest cyclic superstring, i.e. to find a short sequence as possible which contains all the sequences of a given set $\mathbb{S}$ as subsequences. The acyclic version of the SCS problem, which find the shortest superstring, has been extensively studied in the literature. This problem has numerous applications in data compression [49] and computational biology [29]. However, the problem is known to be NP-complete [28] and even APX-complete [4], indicating that polynomial-time approximation schemes with arbitrary constant approximations are not to be expected.

Many greedy and approximation algorithms have been developed to tackle this problem, and they have demonstrated numerical efficiency [34], [35], [48]. Furthermore, we note that any solution for the acyclic SCS problem can also be considered a feasible solution for the cyclic SCS problem. In our variant to the problem we built a complete directed graph which represent all the sequences of the code $\mathcal{C}$ and their possible overlaps with all the other sequences that follow them. For this purpose we use the well-known approximation algorithm for the set cover problem [53, Chapter 2].

While comparing codes with codewords of different lengths is not trivial, it is much easier to compare codes in which all codewords have the same length. Given two (n,m,R)-CSCs, it is obvious that the code with a smaller number of codewords will be considered better. Given two codes, $\mathcal{C}_1$ an (n,m_1,R)-CSC and $\mathcal{C}_2$ an (n,m_2,R)-CSC where the total length of the codewords is smaller in $\mathcal{C}_1$ and $m_1 \geq m_2$, then $\mathcal{C}_1$ is considered superior to $\mathcal{C}_2$. However, if the total length of the codewords in $\mathcal{C}_1$ is smaller and $m_1 < m_2$, then the two codes are incomparable. Nevertheless, there should be some tradeoff between the two parameters. We demonstrate these concepts for a length $n = 2^k - 1$ in this section and for a length $n = 2^k$ in Section IV.

For a length $n = 2^k - 1$ we consider the Hamming code $\mathcal{C}$. The Hamming code $\mathcal{C}$ of length $2^k - 1$ can be represented as a cyclic covering code that contains 2^{2^k-k-1} codewords. If c is a codeword in $\mathcal{C}$ then all the cyclic shifts of c are also codewords in $\mathcal{C}$. Each codeword of $\mathcal{C}$ has d distinct cyclic shifts, where d is a divisor of n. For example, the all-zero word is a codeword and it has one distinct cyclic shift. From the set of d distinct cyclic shifts only one is considered in the code.

Example 5. For $n = 15$, we consider the cyclic Hamming code of length 15. It has 134 codewords of length 15, 6 codewords of length 5, 2 codewords of length 3, and 2 codewords of length 1. Together, these 144 codewords contain 2048 words of length 15 which form the Hamming code of length 15. They are merged to form a $(15,1)$-CS of length 3516 (see Appendix C) compared to a lower bound 2048 obtained by the sphere-covering bound. ■

In general, the Hamming code $\mathcal{C}$ contains 2^{2^k-k-1} codewords. There are at most

$$\sum_{d|n,\ d<n} 2^d < 2^{\lfloor \frac{n}{2} \rfloor + 1} = 2^{2^{k-1}}$$

cyclic words that contain less than n distinct cyclic shifts (see [19, p. 105, Lemma 3.20]). Thus, the $(2^k-1, 2^k-1, 1)$-CSC code contains much less than $\frac{2^{2^k-k-1}}{2^k-1} + 2^{2^{k-1}}$ codewords and hence asymptotically it is an optimal CSC.

We can make a more precise computation when $n = 2^k - 1$ is a prime. In this case only two codewords, the all-zero codeword and the all-one codeword, have less than n distinct shifts. Therefore, the number of codewords in the $(2^k - 1, 2^k - 1, 1)$-CSC is $\frac{2^{2^k-k-1}-2}{2^k-1} + 2$. To obtain a $(2^k - 1, 1)$-CS we have to add to each sequence at most $2^k - 2$ bits to convert it to acyclic sequence which contains all its words of length $2^k - 1$. The $(2^k - 1, 1)$-CS that is generated has length shorter than 2^{2^k-k}, i.e., within factor of at most 2 from optimality.

IV. Construction from Self-Dual Sequences

In this section, we continue with the construction of cyclic codes as presented in Section III by using a specific family of sequences called self-dual sequences. A self-dual sequence is a cyclic sequence $\mathcal{S}$ whose complement $\bar{\mathcal{S}}$ is the same sequence as $\mathcal{S}$ after another shift. The construction will be described for n that it is a power of two, but it can be described to other values of n. For a given word of length n we are interested in self-dual sequences of length $2n$. Such sequences are of the form $[X\ \bar{X}]$, where X is a sequence of length n and $[X\ \bar{X}]$ has no periodicity since its length is a power of 2.

The construction for self-dual sequences starts with a $(n, 2n, 1)$-CSC $\mathcal{C}$, where $n = 2^k$, (with $2^{2^{k+1}-k-2}$ n-tuples), in which for each n-tuple U of a codeword in $\mathcal{C}$ there exists exactly one other n-tuple V of a codeword in $\mathcal{C}$ such that $d(U, V) = 1$. For this purpose, the code $\mathcal{C}$ will be designed in such a way that if $[X\bar{X}]$ is a codeword in $\mathcal{C}$, then there exists exactly one codeword $[Y, \bar{Y}]$ in $\mathcal{C}$ such that $d(X, Y) = 1$. This immediately implies the following observation proved also in [6].

Lemma 6. *The n-tuples in the codeword of $\mathcal{C}$ form an $(n, 1)$-covering code $\mathcal{C}$ for which each codeword $c \in \mathcal{C}$ there exists exactly one other codeword $c' \in \mathcal{C}$ such that $d(c, c') = 1$.*

Now, we are in a position to present a construction for an $(n, 1)$-CS based on self-dual sequences of length $2n$.

Construction 1. Let $\mathcal{C}_n$ be an $(n, 2n, 1)$-CSC code, $n = 2^k$, with 2^{2^k-2k-1} sequences, all of them are self-dual. Moreover, if $c \in \mathcal{C}$ then there exists a shift $[X\ \bar{X}]$ of c for which $\mathcal{C}$ has exactly one other codeword $[X'\ \bar{X}']$, where X' differs from X only in the last coordinate.

Let $\mathcal{E}_n$ be the set of all 2^{n-2} even-weight words in $\mathbb{F}_2^n$ that start with a *zero*. Let $\mathcal{C}_{2n}$ be the code with self-dual sequences of length $4n$ defined by

$$\mathcal{C}_{2n} \triangleq \{[U\ U + X\ \bar{U}\ \bar{U} + X]\ :\ U \in \mathcal{E}_n,\ [X\ \bar{X}] \in \mathcal{C}_n]]\},$$

where the basis is

$$\mathcal{C}_8 = [0001101111100100], [0001101011100101]$$

■

The computation of the sizes of the codes and the correctness of the details can be found in [6], where the following result was proved.

Theorem 7. *The code $\mathcal{C}_{2n}$, $n = 2^k$, of Construction 1 is an $(2n, 4n, 1)$-CSC with $2^{2^{k+1}-2k-3}$ codewords which are self-dual sequences of length 2^{k+1}. If a codeword is in $\mathcal{C}_{2n}$ then it has the form $[X\ \bar{X}]$ and $[X'\ \bar{X}']$ is another codeword in $\mathcal{C}_{2n}$.*

Remark 8. Construction 1 is very similar to the constructions presented in [23], [25]. The construction based on self-dual sequences was found by one of the authors and was motivated by the introduction of nearly-perfect covering codes [6]. It was first introduced for the earlier conference version of this paper [9].

From the two codewords of $\mathcal{C}_8$, we can form the optimal $(8, 1)$-CS of length 32

$$[0001101111100100\ 0001101011100101]\ .$$

Construction 1 is applied on $\mathcal{C}_8$ to obtain a $(16, 32, 1)$-CSC with 128 codewords of length 32. These 128 codewords are partitioned into 64 pairs of the form

$$[X\ \bar{X}] \quad \text{and} \quad [X'\ \bar{X}'] \ .$$

Each pair is combined into one sequence of length 64 that contains all the 16-tuples of the original two codewords of length 32. Hence, this code is an optimal $(16, 64, 1)$-CSC of length 64 with 64 codewords. Trivially, each sequence can be extended with its first 15 bits to an acyclic sequence of length $64 + 15 = 79$. In other words, these 64 sequences can be concatenated to form a $(16, 1)$-CS of length $64 \cdot (64 + 15) = 5056$. However, we can merge these sequences with overlaps between the suffixes and the prefixes of the sequences and obtain a $(16, 1)$-CS of length 4462 (see Appendix D), where the known lower bound, for a $(16, 1)$-covering code, is 4096. The same idea can be applied recursively on the 64 pairs of sequences of length 32 to obtain an upper bound on the shortest length of a $(2^k, 1)$-CS.

For $n = 2^k$, $k > 2$, we obtain a $(2^k, 2^{k+2}, 1)$-CSC of length 2^{k+2} with 2^{2^k-2k-2} codewords. This code is optimal as the total length of the codewords is the same as the number of codewords in an optimal $(2^k, 1)$-covering code.

Without considering any overlap we can merge these 2^{2^k-2k-2} sequences, whose length is $2^{k+2}+2^k-1$ (the $n-1$ prefix is concatenated after the last bit of each sequence) as an acyclic sequence, to obtain a $(2^k, 1)$-CS of length $2^{2^k-2k-2}(2^{k+2} + 2^k - 1)$. We can also compute some overlaps between the prefixes and the suffixes of these sequences; however, we omit this computation due to its complexity. The number of codewords in an optimal $(2^k, 1)$-covering code is $K = 2^{2^k-k}$, so this sequence has a length of less than $1.25K$, i.e., within factor of at most 1.25 from optimality.

V. Interleaving of Covering Sequences

Interleaving is probably the most simple method to construct (n, R)-CSs with a large length and a large covering radius from two or more covering sequences of smaller length and a smaller covering radius. When two covering sequences participate in the construction, one is an (n_1, R_1)-CS of length k_1 and the other is an (n_2, R_2)-covering sequence of length k_2, where $n_2 \leq n_1 \leq n_2 + 1$ and $\gcd(k_1, k_2) = 1$. From these two covering sequences we generate an $(n_1 + n_2, R_1 + R_2)$-CS of length $2k_1k_2$.

Construction 2. Let $\mathcal{A} = [a_0, a_1, \ldots, a_{k_1-1}]$ be an (n_1, R_1)-CS and let $\mathcal{B} = [b_0, b_1, \ldots, b_{k_2-1}]$ be an (n_2, R_2)-CS and assume further that $n_1 = n_2$ or $n_1 = n_2 + 1$ and $\gcd(k_1, k_2) = 1$. The interleaving of $\mathcal{A}$ and $\mathcal{B}$, is the sequence $\mathcal{S} = [s_0, s_1, \ldots, s_{2k_1k_2-1}]$ of length $2k_1k_2$ defined by $s_0 = a_0$, $s_1 = b_0$, $s_2 = a_1$, $s_3 = b_1$, and in general $s_{2i} = a_i$, where i is taken modulo k_1 and $s_{2i+1} = b_i$, where i is taken modulo k_2, for $0 \leq i \leq k_1k_2 - 1$. ■

Theorem 9. *If $n_1 = n_2$ or $n_1 = n_2 + 1$, then the sequence $\mathcal{S}$ defined in Construction 2 is an $(n_1 + n_2, R_1 + R_2)$-CS of length $2k_1k_2$.*

Proof. First note that since $\gcd(k_1, k_2) = 1$, it follows that for each i and j, $0 \leq i \leq k_1-1$, $0 \leq j \leq k_2-1$ the string $a_ib_ja_{i+1}b_{j+1}\cdots a_{i+n_1}b_{j+n_1}a_{i+n_1+1}$, where indices of $\mathcal{A}$ are taken modulo k_1 and indices of $\mathcal{B}$ are taken modulo k_2, is a subsequence in $\mathcal{S}$. This also implies that the length of the sequence is $2k_1k_2$. We distinguish between the two cases of $n_1 = n_2$ and $n_1 = n_2 + 1$.

Case 1: If $n_1 = n_2 = n$, then consider a word $(\alpha_1, \beta_1, \alpha_2, \beta_2, \ldots, \alpha_n, \beta_n)$. Since $\mathcal{A}$ is an (n, R_1)-CS and $\mathcal{B}$ is an (n, R_2)-CS, it follows that there exists a subsequence $a_ia_{i+1}\cdots a_{i+n-1}$ of $\mathcal{A}$ whose distance from $\alpha_1, \alpha_2, \ldots, \alpha_n$ is at most R_1 and there exists a subsequence $b_jb_{j+1}\cdots b_{j+n-1}$ of $\mathcal{B}$ whose distance from $\beta_1, \beta_2, \ldots, \beta_n$ is at most R_2. Therefore, the subsequence of $\mathcal{S}$, $a_ib_ja_{i+1}b_{j+1}\cdots a_{i+n-1}b_{j+n-1}$ is within distance at most $R_1 + R_2$ from $(\alpha_1, \beta_1, \alpha_2, \beta_2, \ldots, \alpha_n, \beta_n)$.

Case 2: If $n_1 = n_2 + 1$, then consider a word $(\alpha_1, \beta_1, \alpha_2, \beta_2, \ldots, \alpha_{n_2-1}, \beta_{n_2-1}, \alpha_{n_2}, \beta_{n_2}, \alpha_{n_1})$. Since we have that $\mathcal{A}$ is an (n_1, R_1)-CS and $\mathcal{B}$ is an (n_2, R_2)-CS, it follows that there exists a subsequence $a_ia_{i+1}\cdots a_{i+n_1-1}$ of $\mathcal{A}$ whose distance from $\alpha_1, \alpha_2, \ldots, \alpha_{n_1}$ is at most R_1 and there exists a subsequence $b_jb_{j+1}\cdots b_{j+n_2-1}$ of $\mathcal{B}$ whose distance from $\beta_1, \beta_2, \ldots, \beta_{n_2}$ is at most R_2. Therefore, the

subsequence of $\mathcal{S}$, $a_i b_j a_{i+1} b_{j+1} \cdots a_{i+n_2-1} b_{j+n_2-1} a_{i+n_1-1}$ is within distance at most $R_1 + R_2$ from the word $(\alpha_1, \beta_1, \alpha_2, \beta_2, \ldots, \alpha_{n_2}, \beta_{n_2}, \alpha_{n_1})$. □

Example 10. The following (n, R)-CSs were obtained by using Construction 2 on two shorter covering sequences which are either trivial or presented in Appendix A or in Appendix B.

Consider the $(9, 0)$-CS of length 512 and a $(9, 1)$-CS length 93. Interleaving these two sequences using Construction 2 yields a $(18, 1)$-CS of length $2 \cdot 512 \cdot 93 = 95232$.

Consider the $(10, 0)$-CS of length 1024 and a $(10, 1)$-CS length 175. Interleaving these two sequences using Construction 2 yields a $(20, 1)$-CS of length $2 \cdot 1024 \cdot 175 = 358400$.

Consider an $(8, 1)$-CS of length 32 and a $(9, 1)$-CS length 93. Interleaving these two sequences using Construction 2 yields a $(17, 2)$-CS of length $2 \cdot 32 \cdot 93 = 5952$.

Consider an $(8, 1)$-CS of length 37 and an $(8, 2)$-CS length 14. Interleaving these two sequences using Construction 2 yields a $(16, 3)$-CS of length $2 \cdot 37 \cdot 14 = 1036$.

Consider an $(8, 1)$-CS of length 37 and a $(9, 2)$-CS length 20. Interleaving these two sequences using Construction 2 yields a $(17, 3)$-CS of length $2 \cdot 37 \cdot 20 = 1480$.

Consider a $(9, 1)$-CS of length 93 and a $(9, 2)$-CS length 20. Interleaving these two sequences using Construction 2 yields a $(18, 3)$-CS of length $2 \cdot 93 \cdot 20 = 3720$.

Consider a $(9, 1)$-CS of length 93 and a $(10, 2)$-CS length 38. Interleaving these two sequences using Construction 2 yields a $(19, 3)$-CS of length $2 \cdot 93 \cdot 38 = 7068$.

Consider a $(10, 1)$-CS of length 175 and a $(10, 2)$-CS length 38. Interleaving these two sequences using Construction 2 yields a $(20, 3)$-CS of length $2 \cdot 175 \cdot 38 = 13300$.

■

Construction 2 can be generalized to three or more sequences. When t sequences are interleaved the requirement is that each sequence is an (n_i, R_i)-CS of length k_i, $1 \leq i \leq t$, where $\gcd(k_i, k_j) = 1$, $1 \leq i < j \leq t$, and $n \leq n_i \leq n + 1$ for some positive integer n. From these t sequences we generate an $(\sum_{i=1}^{t} n_i, \sum_{i=1}^{t} R_i)$-CS sequence of length $\ell = t \prod_{i=1}^{t} k_i$, $[s_0 s_1 \ \cdots \ s_{\ell-1}]$, where s_r, $r \equiv j \pmod t$, $1 \leq j \leq t$, is a bit from the j-th sequence. Unfortunately, the factor t in the length of the sequence makes this construction quite weak for $t > 2$ and also for $t = 2$ (see Construction 2) it is not too effective, but some of our best (n, R)-CSs for $n \leq 20$ and $1 \leq R \leq 3$ are obtained by this construction. Fortunately, we can get rid of the factor 2 when $t = 2$ and $n_1 = n_2$ at the expense of some extra small redundancy as follows.

Construction 3. Let $\mathcal{A} = [a_0, a_1, \ldots, a_{k-1}]$ be an (n, R)-CS in which $a_i = 0$ for $0 \leq i \leq n - 2$, i.e., $\mathcal{A}$ has a subsequence with $n - 1$ consecutive *zeros* (the same can be applied for the *ones*). If k is even then form the following sequence $\mathcal{S}$ presented in $k/2$ parts. The first part is

$$a_0 a_0 a_1 a_1 \ \cdots \ a_{k-1} a_{k-1} a_0 0 \ .$$

The second part is

$$a_1 a_0 a_2 a_1 \ \cdots \ a_{k-1} a_{k-2} a_0 a_{k-1} a_1 0 \ .$$

The i-th part, $1 \leq i \leq k/2$, is

$$a_{i-1} a_0 a_i a_1 a_{i+1} a_2 \ \cdots \ a_{i-2} a_{k-1} a_{i-1} 0 \ .$$

These $k/2$ parts, each one of length $2k+2$, are concatenated together, in their order, to the final sequence $\mathcal{S}$.

If k is odd, then form the same parts, where the last one is for $i = \frac{k+1}{2}$. ■

Theorem 11. *The sequence $\mathcal{S}$ defined in Construction 3 is an $(2n, 2R)$-CS of length $k(k+1)$ if k is even and $(k+1)^2$ if k is odd.*

Proof. The proof is essentially as the one of Theorem 9. The main difference is that the two sequences $\mathcal{A}$ and $\mathcal{B}$ in Construction 2 is replaced with only one sequence in Theorem 9. The sequence $\mathcal{B}$ is replaced

by $\mathcal{A}$ to which a *zero* is concatenated at the end. All the associated shifts between the different positions of the sequence are guaranteed by the different parts. The 0 at the end of each part makes sure that we will move to another shift of $\mathcal{A}$. It does not destroy any n-tuples since it can be considered as adding another *zero* to the run of $n-1$ *zeros* in one of the interleaved sequences which does not damage any of the covered words of length n. □

Example 12. The following $(n, 2)$-CSs were obtained by using Construction 3.

Consider a $(8, 1)$-CS of length 40 with a subsequence of 7 consecutive *zeros* apply Construction 3 to obtain a $(16, 2)$-CS of length $40 \cdot 41 = 1640$.

Consider a $(9, 1)$-CS of length 102 with a subsequence of 8 consecutive *ones* apply Construction 3 to obtain an $(18, 2)$-CS of length $102 \cdot 103 = 10506$.

Consider a $(10, 1)$-CS of length 177 with a subsequence of 9 consecutive *zeros* apply Construction 3 to obtain a $(20, 2)$-CS of length $178 \cdot 178 = 31684$.

■

VI. A Construction from Primitive Polynomials

A construction based on a specific type of primitive polynomials usually does not generate a very short sequence. However, one of our specific bounds (see Table I) comes from such a sequence. Shift-register sequences and especially those which are produced by a linear function or a modification of such function are very useful for various applications and for constructing of other related combinatorial structures [19], [30]. The same approach is taken in this section. Of special interest are binary sequences generated by a linear function and whose length is $2^n - 1$. These sequences are called M-sequences (for maximal length) and their function is derived from a coefficient of a primitive polynomial.

Let $c(x) = \sum_{i=0}^{n} c_i x^i$, where $c_n = c_0 = 1$ and $c_i \in \{0, 1\}$ for $1 \leq i \leq n-1$, be an irreducible polynomial. Define the following infinite sequence $a_0, a_1, a_2, \ldots$, where

$$a_k = \sum_{i=1}^{n} c_i a_{k-i} \tag{1}$$

with the initial nonzero n-tuple $(a_{-n}, a_{-n+1}, \ldots, a_{-1})$. If $c(x)$ is a primitive polynomial, i.e., a polynomial whose roots have order $2^n - 1$, then the sequence $\mathcal{A} = [a_0, a_1, a_2, \ldots]$ is called an M-sequence, its period is period $2^n - 1$, and hence only its first $2^n - 1$ terms are considered. In this cyclic sequence each nonzero n-tuple appears exactly once as a window of length n. Each root of the primitive polynomial generates the field GF(2^n), i.e., it is a primitive element of the field. The $2^n - 1$ consecutive cyclic shifts of $\mathcal{A}$ can be viewed as another isomorphic representation of the field, These details are well-explained in these two books [19], [30].

Consider now another recursion for a sequence defined by

$$b_k = \sum_{i=1}^{n} c_i b_{k-i} + 1 \tag{2}$$

with the initial nonzero n-tuple $(b_{-n}, b_{-n+1}, \ldots, b_{-1})$.

Lemma 13. *The sum $\sum_{i=1}^{n} c_i$ is an even integer.*

Proof. This follows from the fact that after n consecutive *ones* we should have a *zero* in such a sequence. Another argument is that if $\sum_{i=1}^{n} c_i$ is an odd integer, then $\sum_{i=0}^{n} c_i = 0$, 1 is a root of $c(x)$ and hence $c(x)$ is not a primitive polynomial. □

Lemma 14. *The sequence $\mathcal{B} = (b_0, b_1, b_2, \ldots)$ is the binary complement of the sequence $\mathcal{A}$.*

Proof. Let $\mathbf{a} \triangleq (a_j, a_{j+1}, \ldots, a_{j+n-1})$ be an n-tuple in the sequence $\mathcal{A}$. By Equation (1) we have that

$$a_{j+n} = \sum_{i=1}^{n} c_i a_{j+n-i}.$$

Let $\mathbf{b} \triangleq (b_j, b_{j+1}, \ldots, b_{j+n-1})$ be an n-tuple in the sequence $\mathcal{B}$, where $b_i = a_i + 1$, for $j \leq i \leq j+n-1$, i.e., $\mathbf{b} = \bar{\mathbf{a}}$. Therefore, we have that

$$\begin{aligned} b_{j+n} &= \sum_{i=1}^{n} c_i b_{j+n-i} + 1 = \sum_{i=1}^{n} c_i (a_{j+n-i} + 1) + 1 = \sum_{i=1}^{n} c_i a_{j+n-i} + \sum_{i=1}^{n} c_i + 1 \\ &= \sum_{i=1}^{n} c_i a_{j+n-i} + 1 = a_{j+n} + 1 \end{aligned} .$$

This implies that the sequence $\mathcal{B}$ is the binary complement of the sequence $\mathcal{A}$. □

Corollary 15. *The recursion of Equation* (1) *generates the sequence $\mathcal{A}$ and the all-zero sequence. The recursion of Equation* (2) *generates the sequence $\mathcal{B} = \bar{\mathcal{A}}$ and the all-one sequence.*

Lemma 16. *Let $c(x) = \sum_{i=0}^{n} c_i x^i$ be a primitive polynomial for which $c_i = 0$ for $1 \leq i \leq 2R+1$ and consider the sequences $\mathcal{A}$, $\mathcal{B}$, the all-zeros sequence and the all-ones sequence. The code that contains these four sequences is an $(n + 2R + 1, R)$-CSC.*

Proof. Let X be any given n-tuple. Since the first $2R+1$ c_is (except for c_0) are *zeros*, it follows that the last $2R+1$ elements of X are not influencing the result of the next bit for both recursions and hence the addition of the 1 in the sequence $\mathcal{B}$ implies that the next $2R+1$ bits after X in $\mathcal{A}$ and $\mathcal{B}$ will be complementing. This implies that $Xz_1z_2 \cdots z_{2R+1}$ is in the ball of radius R either by an $(n+2R+1)$-tuple of $\mathcal{A}$ that starts with X or by an $(n+2R+1)$-tuple of $\mathcal{B}$ that starts with X. □

The number of sequences of the code defined in Lemma 16 is four and they can be efficiently concatenated to one $(n + 2R + 1, R)$-CS.

Theorem 17. *The four sequences of the $(n+2R+1, R)$-CSC can be combined to a $(n+2R+1, R)$-CS of length $2^{n+1} + 2n + 8R + 2$.*

Proof. The span n M-sequence $\mathcal{A}$ has length $2^n - 1$ and the same length has the sequence $\mathcal{B}$. The sequence $\mathcal{A}$ has a subsequence of $n-1$ consecutive *zeros* and hence it can be combined with the all-zero sequence to an acyclic sequence of length $2^n + n + 4R + 1$. The same can be done for the sequence $\mathcal{B}$ and the all-one sequence and the two sequences can be combined with no overlaps to an $(n + 2R + 1, R)$-CS of length $2^{n+1} + 2n + 8R + 2$. □

Remark 18. The sequences generated in Theorem 17 can be combined with some overlap to reduce its length, but this will make only a very small improvement in the length of the covering sequence. It should be noted the required primitive polynomials exist for relative small n and larger one.

Finally, in Table I the current best lower and upper bounds on $\mathcal{L}(n, R)$ are presented, where the lower bounds are either by computer search for very small n or the known lower bound on the smallest size of an (n, R)-covering code.

VII. Folding of a Covering Sequence into a 2D-Sequence

After we have looked at covering sequences and covering sequence codes we continue with a generalization of the one-dimensional framework into two-dimensional arrays. The ultimate goal is to construct an $M \times N$ array $\mathcal{A}$ in which for each $m \times n$ matrix $\mathcal{B}$ there exists an $m \times n$ submatrix $\mathcal{X}$ of $\mathcal{A}$ such that $d(\mathcal{B}, \mathcal{X}) \leq R$. Two techniques will be presented in this section. The first one is a folding technique and the second one involves different related shifts of a one-dimensional covering sequence.

n	$R=1$	$R=2$	$R=3$
9	62-93[a]	20[b]	12[b]
10	107-175[a]	38[b]	16[b]
11	180-283[a]	38-111[a]	20[b]
12	342-597[a]	62-161[a]	34-40[b]
13	598-1172[a]	97-292[a]	34-93[a]
14	1172-2271[a]	159-525[a]	44-239[c]
15	2048-3516[e]	310-907[a]	70-406[a]
16	4096-4462[f]	512-1640[c]	115-1036[d]
17	7419-17719[a]	859-5952[d]	187-1480[d]
18	14564-95232[d]	1702-10506[c]	316-3720[d]
19	26309-176170[g]	2898-31684[h]	513-7068[d]
20	52618-358400[d]	5330-31684[c]	892-13300[d]

Table I: Bounds for the length of the shortest (n,R)-CS for $9 \le n \le 20$ and $R=1,2,3$.
a - computer search, b - Chung and Cooper [10], c - Construction 3, d - Construction 2, e - from the Hamming code, f - from self-dual sequences, g - from primitive polynomial, h - Theorem 3.

For the first technique, folding, we will use the following simple lemma that was already observed in [10].

Lemma 19. *If there exists an (n,R)-CS of length k, then there exists an (n,R)-CS sequence of length $k+n-1+\epsilon$ for any $\epsilon \ge 0$.*

Proof. If the (n,R)-CS $\mathcal{S}$ starts at a certain position, then we can append to $\mathcal{S}$ the first $n-1$ bits and any ϵ bits after that to keep it an (n,R)-CS. □

Construction 4. Start with an (mn,R)-CS $\mathcal{S} = s_0, s_1, \ldots, s_{k-1}$ of length k, where without loss of generality k is divisible by n (see Lemma 19, where $0 \le \epsilon < n-1$). The sequence $\mathcal{S}$ is folded row by row into an $M \times n$ array $\mathcal{A}$, where $M = \frac{k}{n}$. The array is extended into an $M \times (2n-1)$ array $\mathcal{A}'$ by adding to each row of the $M \times n$ array $\mathcal{A}$ the next $n-1$ symbols from $\mathcal{S}$ associated with the given row. In other words, the j-th row of the array $\mathcal{A}'$, where $0 \le j \le \frac{k}{n}-1$, is defined by

$$s_{jn+1}, s_{jn+2}, \ldots, s_{jn+n}, s_{jn+n+1}, \ldots, s_{jn+2n-1}, \tag{3}$$

where the indices are taken modulo k. ■

Before proving that $\mathcal{A}'$ generated in Construction 4 is an $(m \times n, R)$-C2DS we will prove one property of the array $\mathcal{A}'$.

Lemma 20. *The array $\mathcal{A}'$ generated in Construction 4 contains the $m \times n$ subarray*

$$\begin{array}{cccc} s_{jn+i} & s_{jn+i+1} & \cdots & s_{jn+i+n-1} \\ s_{(j+1)n+i} & s_{(j+1)n+i+1} & \cdots & s_{(j+1)n+i+n-1} \\ \vdots & \vdots & \ddots & \vdots \\ s_{(j+m-1)n+i} & s_{(j+m-1)n+i+1} & \cdots & s_{(j+m-1)n+i+n-1}, \end{array}$$

where $0 \le i \le n-1$, $0 \le j \le \frac{k}{n}-1$, and indices are taken modulo k.

Proof. This is an immediate consequence of Equation (3) which defines the j-th row of the array $\mathcal{A}'$. □

Corollary 21. *Each $m \times n$ matrix obtained by folding mn consecutive bits of $\mathcal{S}$, row by row, is contained as an $m \times n$ subarray of $\mathcal{A}'$.*

Theorem 22. *The array $\mathcal{A}'$ generated in Construction 4 is an $(m \times n, R)$-C2DS.*

Proof. Given an $m \times n$ matrix $\mathcal{B}$ we have to show that there exists an $m \times n$ subarray $\mathcal{X}$ of $\mathcal{A}'$ such that $d(\mathcal{B},\mathcal{X}) \le \mathcal{R}$. Let $X_1, X_2, \ldots, X_m$ be the m consecutive rows of $\mathcal{B}$ and let $\mathcal{T} = X_1 X_2 \cdots X_m$ the

sequence of length mn obtained by concatenating them. The sequence $\mathcal{T}$ has length nm and hence there exists a subsequence $\mathcal{Y}$ in $\mathcal{S}$ such that $d(\mathcal{T},\mathcal{Y}) \leq R$. By corollary 21 the $m \times n$ matrix $\mathcal{Y}'$ obtained by folding $\mathcal{Y}$ into an $m \times n$ matrix is contained in $\mathcal{A}'$. Since $d(\mathcal{T},\mathcal{Y}) \leq R$ it follows that $d(\mathcal{B},\mathcal{Y}') \leq R$ which completes the proof. □

Construction 4 implies the following consequence.

Theorem 23. *If there exists an (mn, R)-CS of length k, where n divides k, then there exists a $(m \times n, R)$-C2DS of size $M \times N$, where $M = \frac{k}{n}$ and $N = 2n - 1$.*

If there exists an (mn, R)-CS of length k, where n does not divide k, then there exists an $(m \times n, R)$-C2DS of size $M \times N$, where $N = 2n - 1$ and $\frac{k}{n} \leq M \leq \lceil \frac{k}{n} \rceil + m$.

The main disadvantage of Construction 4 is that the width of the array $N = 2n-1$ is not large enough compared to the width of the window n. Next, we will be interested in starting with an (mn, R)-CS sequence $\mathcal{S} = [s_0, s_1, s_2, \ldots, s_{k-1}]$ and construct an $M \times N$ $(m \times n, R)$-C2DS for which M will be considerably larger than m and N considerably larger than n and the ratio $\frac{MN}{k}$ should be as small as possible (i.e., with as little redundancy as possible). There are a few ways to construct such an array using the same principles as in Construction 4. The sequence is partitioned into r subsequences which form an (mn, R)-CSC. The r sequences should be of the same length κ which is divisible by n. From each sequence we create an $M \times (2n-1)$ array and we concatenate these r arrays into one $M \times (2nr-r)$ array which is a $(m \times n, R)$-C2DS. The specific details are omitted as they are the same as in Construction 4.

The folding construction is very simple and asymptotically, it yields optimal $(m \times n, R)$-C2DS if the (mn, R)-CS is asymptotically optimal. As we mentioned, Proposition 2 can be obtained by using the folding construction, along with the covering sequence whose existence was proved by Vu [54]. The length of this (mn, R)-CS is close up to a factor of $\log mn$ from the sphere-covering bound.

It is natural to think that an (mn, R)-CS of length k_1, will have a smaller length than the area of an $(m \times n, R)$-C2DS with dimension $M \times N$, i.e., $k_1 \leq M \cdot N$. This is indeed the case when folding is applied, $M \cdot N$ is about twice as large as k_1. However, we provide one more construction for covering 2D-sequences from covering sequences, whose outcome is quite surprising as it produces smaller arrays (in area) than the length of the ones obtained by folding the related sequence. Moreover, sometimes it yields an array whose area is smaller than the associated best known covering sequence. This second technique is done by considering different shifts of an one-dimensional (n, R)-CS. It will be presented in two constructions, where the first one is very simple.

Construction 5. Let $\mathcal{S}$ be an (n, R)-CS of length k.

If k is even, then form an $(k+1) \times k$ array whose i-th row, $0 \leq i \leq k-1$ is $\mathbf{E}^j S$, where $j = \sum_{\ell=0}^{i} \ell$, $\mathbf{E}^j S$ is a cyclic shift of $\mathcal{S}$ by j positions to the left, i.e.,

$$\mathbf{E}^j[s_0, s_1, \ldots, s_{k-1}] = [s_j, s_{j+1}, \ldots, s_{k-1}, s_0, \ldots, s_{j-1}],$$

and the k-th row is the same as the $(k-1)$th row.

If k is odd, then form an $k \times k$ array whose i-th row, where $0 \leq i \leq k-1$ is $\mathbf{E}^j S$, $j = \sum_{\ell=0}^{i} \ell$. ■

Theorem 24. *If n is even, then the $(k+1) \times k$ array obtained in Construction 5 is an $(2 \times n, 2R)$-C2DS. If n is odd, then the $k \times k$ array obtained in Construction 5 is an $(2 \times n, 2R)$-C2DS.*

Proof. Let $\mathcal{A}$ a $k \times k$ array obtained in Construction 5 and let $\mathcal{B}$ a $2 \times n$ matrix that consists of two sequences (rows) of length n, the first one X and the second one Y. Each row of $\mathcal{A}$ is the sequence $\mathcal{S}$ at some shift, where the sequence $\mathcal{S}$ of the i-th row, $1 \leq i \leq k$, is shifted by i positions to the left compared to the sequence in the $(i-1)$-th row. The 0-th row is taken without shifting.

Assume first that k is even. Since $\mathcal{S}$ is an (n, R)-CS, it follows that there exists a subsequence U of length n in $\mathcal{S}$ such the $d(X, U) \leq R$ and a subsequence V of length n in $\mathcal{S}$ such that $d(Y, V) \leq R$. Since $\mathcal{S}$ is an (n, R)-CS and all possible shifts are taken between consecutive rows, it follows that in two

consecutive rows of $\mathcal{A}$ we have the subsequence U and V one on top of the other, i.e., a $2 \times n$ matrix $\mathcal{W}$ such that $d(\mathcal{W}, \mathcal{B}) \leq 2R$. Thus, the $(k+1) \times k$ array obtained in Construction 5 is an $(2 \times n, 2R)$-C2DS.

When k is odd the proof is similar, but we have to notice that since $\sum_{i=1}^{k} i \equiv 0 \pmod{k}$, it follows that the 0-th row is without any shift compared to the last row and hence one less row is required. □

Example 25. Consider the $(6, 1)$-CS of length 12 presented in Appendix A. By applying the defined shifts of Construction 5 on this sequence which is the first row in the array we obtain a 13×12 $(2 \times 6, 2)$-C2DS whose area is 156. The best associated $(12, 2)$-CS that we find is based on computer search has length 161 (see Appendix B). The generated 13×12 $(2 \times 6, 2)$-C2DS whose area is 156 which smaller than the related $(12, 2)$-CS of length 161 that was found by computer search. It is the following array:

$$\begin{array}{cccccccccccc}
0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\
0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\
1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\
1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\
1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \\
1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\
0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\
0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\
0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\
1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\
1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\
1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0
\end{array}$$

■

Example 26. Consider the $(7, 1)$-CS of length 22 presented in Appendix A. By applying the defined shifts of construction 5 on this sequence which is the first row in the array, we obtain a 23×22 $(2 \times 7, 2)$-C2DS whose area 506. The best related $(14, 2)$-CS that we found is based on computer search has length 525 (see Appendix B).

■

Finally, Construction 5 can be generalized as follows.

Construction 6. Let $\mathcal{S}$ be an (n, R)-CS of length k and let $\mathcal{T} = [t_1, t_2, \ldots, t_{k^{m-1}}]$ be a span $m-1$ de Bruijn sequence over Σ_k. We form the following $k^{m-1} \times k$ array $\mathcal{A}$, where the sequence $\mathcal{S}$ in the i-th row of $\mathcal{A}$, $1 \leq i \leq k^{m-1}$ is shifted t_i positions related to the sequence $\mathcal{S}$ in the $(i-1)$-th row, where the 0-th row is considered to be with no shift. ■

Theorem 27. *The $k^{m-1} \times k$ array $\mathcal{A}$ of Construction 6 is an $(m \times n, mR)$-C2DS.*

Proof. The proof is very similar to the one of Theorem 24. First note that the sum of all the shifts is zero and hence the virtual 0-th row can be considered to be with no shifts. For any $m \times n$ matrix $\mathcal{B}$ we have to find an $m \times n$ window $\mathcal{X}$ in $\mathcal{A}$ such that $d(\mathcal{B}, \mathcal{W}) \leq mR$. Let $B_1, B_2, \ldots, B_m$ be the m consecutive rows of $\mathcal{B}$. Each B_i has length n and hence there exists a subsequence X_i of length n in $\mathcal{S}$ such that $d(B_i, X_i) \leq R$. Let $(i_1, i_2, \ldots, i_{m-1})$ be the $m-1$ consecutive shifts of $\mathcal{S}$ such that these m copies of $\mathcal{S}$ placed on each other contain $\mathcal{B}$ as a window. Since $(i_1, i_2, \ldots, i_{m-1})$ is an $(m-1)$-tuple over $\mathbb{Z}_k$ and $\mathcal{T}$ is a span $m-1$ de Bruijn sequence over $\mathbb{Z}_k$ which implies that $(i_1, i_2, \ldots, i_{m-1})$ is contained in $\mathcal{T}$, it follows that the associated shifts of $\mathcal{S}$ are contained in m consecutive rows of $\mathcal{A}$. These shifts contain an $m \times n$ window $\mathcal{X}$ whose consecutive rows are $X_1, X_2, \ldots, X_m$. Since $d(X_i, B_i) \leq R$, it follows that $d(\mathcal{B}, \mathcal{X}) \leq mR$.

Thus, the $k^{m-1} \times k$ array $\mathcal{A}$ of Construction 6 is an $(m \times n, mR)$-C2DS. □

VIII. Conclusion and Future Research

Covering sequences and covering sequence codes which generalize the well-known covering codes, were considered. Some new construction methods for such covering sequences as well as covering sequence codes were presented. In particular, nearly optimal and asymptotically optimal sequences and codes were obtained using the Hamming codes and self-dual sequences. Finally, generalization for covering 2D-sequences was also discussed and related constructions were given.

This area is far from being fully explored and we conclude with the following problems for future research.

1) The current work and also all the previous papers that considered this topic, have concentrating on the upper bounds of (n, R)-CSs. The lower bounds mentioned in this paper are either the ones used for (n, R)-covering codes or obtained by computer search. We would like to see improvements in these lower bounds and not just by one or two (something which is not difficult to do).
2) Many of the upper bounds used in Table I are quite weak as Construction 2 is used and we do not have for these parameters a stronger construction like Construction 3. We would like to see some new constructions that will make the table more balanced. Similarly, we want to see upper bounds on the sizes of covering sequence codes.
3) The $(2^k - 1, 1)$-CS and the $(2^k, 1)$-CS introduced in Sections III and IV are the best covering sequences obtained for small radii. We would like to have a more precise computation on the length of these sequences. We would also like to see similar sequences for radius 2 and radius 3. The Preparata code used to obtain covering codes for radii 2 and 3 [22], [24] might be the ones to use for this purpose.
4) We would like to see more constructions as well as lower bounds on the size of covering 2D-sequences.
5) We would like to see a more comprehensive study on covering sequence codes, mainly (n, m, R)-CSCs, as the current work is mainly on covering sequences.
6) In the same way that covering codes are defined on other metrics, different from the Hamming that was discussed in the paper, covering sequences can be defined for these metrics. For example, it would be interesting to have such short sequences for various poset spaces.

Recently, Rosin [44] develop a method to search for combinatorial structures. His search found many new such structures including some better covering sequences. The sequences themselves are not yet well documented and we hope he will elaborate on this and improves some of our bounds soon.

Appendix A
Very small (n, R)-CSs used for other bounds

An $(8, 1)$-CS sequence of length 32 - $[00011011111001000001101011100101]$.
An $(8, 1)$-CS sequence of length 35 - $[00010110110111000010001111011101001]$.
An $(8, 1)$-CS sequence of length 37 - $[0001101111100100000110101110011100101]$.
An $(8, 1)$-CS sequence of length 40 (with 7 consecutive *zeros*) - $[0001101111100100000001000001101011100101]$.
An $(8, 2)$-CS sequence of length 14 - $[00111011010010]$.
An $(9, 2)$-CS sequence of length 20 - $[00010010001110110111]$.

Appendix B
Small (n, R)-CSs

The following thirteen codewords form a $(10, 11, 1)$-CSC:

$$[00001010000], [00101001011], [10100101111], [10110111001], [11011101111],$$
$$[11011101111], [01110111100], [11110010011], [11110010000], [11001000101],$$
$$[00100011000], [01000110101], [10001101001], [00011010000]\,.$$

Their extension by nine bits and their associated overlaps are as follows:

00001010000000010100	7	01110111100011101111	4	01000110101010001101	8
00101001011001010010	7	11110010011111100100	9	10001101001100011010	8
10100101111101001011	4	11110010000111100100	7	00011010000000110100	2
10110111001101101110	7	11001000101110010001	7		
11011101111110111011	7	00100011000001000110	8		

The total number of bits in the thirteen sequences is 260 and the total number of overlaps is 85. This yields a $(10, 1)$-CS of length $260 - 85 = 175$:

$$
\begin{aligned}
&[00001010000000010100101100101001011111010010\\
&11011100110110111011111101110111100011101111\\
&00100111111001000011110010001011100100011000\\
&00100011010101000110100110001101000000001101] .
\end{aligned}
$$

The following thirteen codewords form a $(10, 11, 1)$-CSC:

$$
\begin{aligned}
&[11010111111], [01011110000], [10111100101], [11100110011], [11001100001]\\
&[00110001101], [10001101100], [11011010101], [01101010010], [10101000011],\\
&[10000010010], [00000100000], [00001000101] .
\end{aligned}
$$

Their extension by nine bits and their associated overlaps are as follows:

11010111111110101111	7	00110001101001100011	6	10000010010100000100	8
01011110000010111100	8	10001101100100011011	5	00000100000000001000	8
10111100101101111001	6	11011010101110110101	7	00001000101000010001	1
11100110011111001100	8	01101010010011010100	7		
11001100001110011000	7	10101000011101010000	5		

The total number of bits in the thirteen sequences is 260 and the total number of overlaps is 83. This yields a $(10, 1)$-CS of length $260 - 83 = 177$ with a subsequence having 10 consecutive *zeros*:

$$
\begin{aligned}
&[11010111111110101111000001011110010110111100\\
&11001111100110000111001100011010011000110110\\
&01000110110101011101101010010011010100001110\\
&10100000100101000001000000000010001010000100] .
\end{aligned}
$$

The following twenty codewords form a $(11, 11, 1)$-CSC:

$$
\begin{aligned}
&[00111011011], [01110110100], [10110101100], [10101100010], [10001001011],\\
&[10001001010], [01010100111], [01010011010], [01001101110], [10111111110],\\
&[10111111111], [11111111001], [11111000010], [11100001100], [00011011001],\\
&[01100000000], [00000010000], [00001000001], [00000111101], [00011110011] .
\end{aligned}
$$

Their extension by ten bits and their associated overlaps are as follows:

0011101101100111011101	9	0101001101001010011101	8	0001101100100011011100	5
0111011010001110111010	7	0100110111001001101111	5	0110000000001100000000	6
1011010110010110101110	7	1011111111010111111111	10	0000001000000000001000	8
1010110001010101110001	5	1011111111111011111111	8	0000100000100001000000	5
1000100101110001001011	10	1111111100111111111100	7	0000011110100000111110	8
1000100101010001001011	4	1111100001011111001001	8	0001111001100011111001	3
0101010011101010100111	8	1110000011001110000110	6		

The total number of bits in the twenty sequences is 420 and the total number of overlaps is 137. This yields a $(11, 1)$-CS of length $420 - 137 = 283$:

$$
\begin{aligned}
&[00111011011001110110100011101101011001011010110001010101\\
&10001001011100010010101000100101010011101010100110100101\\
&00110111001001101111111101011111111111011111111100111111111\\
&10000101111100001100111000011011001000110110000000001100\\
&00000100000000001000001000010000011110100000111100110001111] .
\end{aligned}
$$

The following sequences is a $(12, 1)$-CS of length 597:

$$
\begin{aligned}
&[10101100111011010110011111001011001111111110110011111111110110\\
&11111111100001101111110000101000011000010100111100001010010 0\\
&00100101001000111100100100011101000110001110100110110111010 0\\
&11011000011001101100010010111011000100011001100010001001100 00\\
&10001001000101000100100101010110010010101000001001010100010 1\\
&01011010001010111001000101011010100010101111110001010111110 1\\
&11101011111010001101111101001011110110100101110001111010111 0\\
&00110110111100011011100111001101110010100100111001010000011 0\\
&01001000001100000000000110000011110011000001111000000110111 1\\
&00000001111110000000101110101000010111010100111011101010 0] .
\end{aligned}
$$

The following sequences is a $(13, 1)$-CS of length 1172:

$$
\begin{bmatrix}
10111001111101101110011111111001110011111100111100111111110011\\
01001011110011010001110000110100011101011000000111010110100\\
10111010110101010000101101010101110011010101011111000000010\\
11111000010000111110000100010010100001000100010010010001000\\
10101000001000101010010110001010100101000110101001010000011\\
11001010000010001101000000100011001110001000110010100110001\\
10010101101101100101011001110101010110011101101001100111011\\
00100000111011001001111010110010011111001100100111101111101\\
00111101111111000111011111110100110111111101111101111111011\\
01010111111011010111011101101011100000011010111000101110 10\\
11100010000110111000100000001110001000000000011000000000000\\
11111101010000111111010010001111110100011111111101000101011\\
11101000101011101010001010111011000011011110110001100111 10\\
11000100101100110001001010010000010010100110100100101001110\\
11110001001101111100001011101111000110101011110001101100010\\
10001101100001110001011000011100100110000111001111100001110\\
01000000001110010001100011100100001010111001000010011011010\\
00010011011100000100110111011001001101110100011011011101000\\
10110001101000101100101100001011001011011111110010110111101\\
00010110111101001100101111010011100001010100111000 0]\,.
$$

The following sequence is a $(14, 1)$-CS of length 2271:

$$
\begin{bmatrix}
11110111101011011110111101010111110111101010010110111101010\\
01000011110101001000110001111001000110001010111000110001010\\
01000011000101001110010001011001110010001010101110010001010\\
11011001000101010000101000101010000111100101010000110100010\\
00100011010001000001101010001000001000000001000001000100001\\
00000100011111100000100011101110000100011101011000100011101\\
01110010001110101010010001110100001001101110100001000000010\\
10000100000111111101100000111111100101011111111100101000001\\
11110010100011001110010100011001000100100011001000000000011\\
00100000010101100100000010011100100000010000110000000010000\\
11011000001000011011101111000011011101010101011011101010100\\
10111110101010010100110101010010100110010110010100110011110\\
00010011001111001001001001111001001111100111001001111111101\\
01100111111111011100011111111011111101111111011111111000010010\\
11111110001101001111110001101001011010001101001010111101101\\
00101011110001000101011110001011101011110001011010111110001\\
01101011000010101101011000110001101011000110010010111000110\\
01001001000011001001001100000101001001100001000011001100001\\
00011110110000100011011111100100011011111010010101011111010\\
01000010111101001000010010011001000010010100101000010010101\\
01000001001010100111011011010100111011100110100111011011000\\
00011101110100011110101110100011100001110100011100111010000\\
01110011101011100110011101011111000011101011111001111101011\\
11100110101001111100110101110100000110101110100110000101110\\
10011010110111010011011111111101001101110011101001101110010 1\\
11001101110010111101001110010111100111100010111100111101100\\
01110011110110110010011110110110010111011010110010111011000\\
10001011101100001011001101100001011011000101001011011000110\\
00101101100011111001101100011111011001111011111011001110110\\
01101100111011010101100111011001011010111011001011011011011\\
00101101001001100101101000100110101101000100111010101000100\\
11100011100010011100000010010011100000010101001100000010101
\end{bmatrix}.
$$

$$
\begin{array}{l}
[10101001001010110101000101100000101000101100000000101010110 \\
0000010111000100000010111001100011101111001100011111011 0011 \\
0001111000010110001111000000001001111000000011110111000000 0 \\
1110011010100001110011011010011110011011010001010011011 0100 \\
0000001101101000011011111101000011011011111000011011011 1000 \\
0001101101110000111101101110000110000001110000110000011 0001 \\
0011000001100001011000001 1000] .
\end{array}
$$

The following six codewords form a $(11, 15, 2)$-CSC:

$$
\begin{array}{l}
[110011101001001], [111010111001011], [110101110000000], \\
[010111001000000], [100101000111101], [001010001101100] .
\end{array}
$$

Their extension by ten bits and their associated overlaps are as follows:

1100111010010011100111010	6	1101011100000001101011100	8	1001010001111011001010001	9
1110101110010111110101110	9	0101110010000000101110010	5	0010100011011000010100011	2

The total number of bits in the six sequences is 150 and the total number of overlaps is 39. This yields a $(11, 2)$-CS of length $150 - 39 = 111$:

$$
\begin{array}{l}
[11001110100100111001110101110010111110101110000000110101 \\
11001000000010111001010001111011001010001101100000101000]
\end{array}
$$

The following nine codewords form a $(12, 13, 2)$-CSC:

$$
\begin{array}{l}
[1000101000010], [0100000000100], [1000000001101], [0000110101101], [1010110011110], \\
[1001110110000], [1000111001001], [1100101111100], [1100101111111] .
\end{array}
$$

Their extension by eleven bits and their associated overlaps are as follows:

10001010000101000101000	6	00001101011010000110101 1	6	10001110010011000111001 0	6
01000000001000100000000 1	10	10101100111101010110011 1	6	11001011111001100101111 1	11
10000000011011000000001 1	6	10011101100001001110110 0	3	11001011111111100101111 1	1

The total number of bits in the nine sequences is 216 and the total number of overlaps is 55. This yields a $(12, 2)$-CS of length $216 - 55 = 161$:

$$
\begin{array}{l}
[10001010000101000101000000001000100000000110110000000000110101 \\
10100001101011001111010101100111011000010011101100011100 1001 \\
10001110010111110011001011111111100101111] .
\end{array}
$$

The following sixteen codewords form a $(13, 13, 2)$-CSC:

$$
\begin{array}{l}
[1111111001011], [1001010011010], [0101001101101], [0011011000110], \\
[0110001100101], [0011001011011], [0101101000001], [1011010000011], \\
[0001001001111], [0100111010111], [1011111110111], [1111011100000], \\
[1110000010001], [1000001000000], [0000000111001], [0111000100001] .
\end{array}
$$

Their extension by twelve bits and their associated overlaps are as follows:

111111100101111111100101	6	010110100000101011010000	11	111000001000111100000 1000	10
100101001101010010100 1101	10	101101000001110110100000 1	4	100000100000010000010000	5
010100110110101010011 0110	8	000100100111100010010011 1	7	000000011100100000001 1100	6
001101100011000110110 0011	8	010011101011101001110101 1	4	011100010000101110001 0000	0
011000110010101100011 0010	8	101111111011101111111011	7		
001100101101100110010 1101	7	111101110000011110111 0000	7		

The total number of bits in the sixteen sequences is 400 and the total number of overlaps is 108. This yields a $(13, 2)$-CS of length $400 - 108 = 292$:

$$
\begin{array}{l}
[1111111001011111111100101001101010010100110110101010011011 00 \\
0110001101100011001010110001100101101100110010110100000 10101 \\
1010000011101101000001001001111000100100111010111010011101 01 \\
1111110111101111111011100000111101110000010001111000001000 00 \\
0100000100000001110010000000111000100001011100010000] .
\end{array}
$$

The following sequence is a $(14, 2)$-CS of length 525:

$$
\begin{array}{l}
[0010111101001000010111101001111111011101001111111011111 00111 \\
1111011000110011111011000110110110001000110110110000011111011 \\
0110000111000110110000111001001101000111001001110011000 10100 \\
1110011001010110010011001010111011111101010111011110000 00101 \\
1011110000000100011110000000110011010010000110011010100 10011 \\
0011010100100101101010100100101101010111100101101010110011 01101 \\
0111010110011010111011110110101111011101001011110011101 0010 \\
1110111000000101110111001010001010111001010001000000111 01000 \\
1000000001100001000000001100100010100000110010] .
\end{array}
$$

The following sequence is a $(15, 2)$-CS of length 907:

$$
\begin{aligned}
&[000010111100110000001011110011001000010111001100100000101110\\
&110010000010100110001000001010010000000000101001000011101000\\
&000100001110100101001000111010010100111011101111010011101110\\
&111010011110111011101001000001110110100100000110010001010000\\
&011001001010000001100100101010110011110010101011001011110111\\
&101100101111010110001111111110101100011010001010110001101001 1\\
&100101101101001110010011110001111001001111000010010100111100\\
&001001101111110000100110111110100011111011111010001011011101\\
&101000101101110011110011000111001111001101110011111100110111\\
&000100010011011100010100101101010001010010110011010001001011\\
&001101101001101100110110100001100011011010000110111001011100\\
&011011100101100000011110010110000011100001111000001110000011\\
&001111111000001100110000100000110011000101100011001100010101\\
&111100110001010100111010000101010011101011100101001110101111\\
&111011111010111111110110101110101010011010111010101100000011 1\\
&0101011] .
\end{aligned}
$$

The following five codewords form a $(13, 13, 3)$-CSC:

$$[0110111110111], [1111101100010], [0110001101000], [1101000001001], [0100000100000] .$$

Their extension by twelve bits and their associated overlaps are as follows:

011011111011101101111011	8	011000110100001100011010 0	6	010000010000001000001000 0	1
111110110001011111010001	7	110100000100111010000010 0	10		

The total number of bits in the five sequences is 125 and the total number of overlaps is 32. This yields a $(13, 3)$-CS of length $125 - 32 = 93$:

$$
\begin{aligned}
&[011011111011101101111101100010111110110001101000011000110100\\
&000100111010000010000001000001000] .
\end{aligned}
$$

The following ten codewords form a $(14, 15, 3)$-CSC:

$$
\begin{aligned}
&[110011000000010111001100000], [011010101001111], [010011101011110], [110101111000100],\\
&[000101100001001], [101101011111101], [111111110001101], [000111011001010],\\
&[001001000101000], [010100001100010] .
\end{aligned}
$$

Their extension by thirteen bits and their associated overlaps are as follows:

1100110000000101100110000000	1	000101100001001000101100001 0	2	001001000101000001001000101 0	5
011010101001111010101010011	6	101101011111101101101011111 1	6	010100001100010010100001100 0	0
010011101011110010011101011 1	8	111111110001101111111110001 1	5		
110101111000100110101111000 1	4	000111011001010000111011001 0	4		

The total number of bits in the ten sequences is 280 and the total number of overlaps is 41. This yields a $(14, 3)$-CS of length $280 - 41 = 239$:

$$
\begin{aligned}
&[111001001110101111000100110101111000101100001001000101100001\\
&011010111111011011010111111111000110111111111000111011001010 0\\
&001110110010010001010000010010001010000110001001010000110 00] .
\end{aligned}
$$

The following sequence is a $(15, 3)$-CS of length 406:

$$
\begin{aligned}
&[100000010100101100000010100100000100010100100000100000010110\\
&010100000010110010001111010110010001111110001110101111110001\\
&110111110101111110111110101111100011001101111100011000110111\\
&100011000110110001011101110110001011101101111001111101101111\\
&001101010000011001101010000011100110010000011100110111000010\\
&100110111000010100000000000010100000001110100100000001110101\\
&101001111110101101001111100110010101111100 1100] .
\end{aligned}
$$

APPENDIX C
$(15, 1)$-CS FROM THE HAMMING CODE OF LENGTH 15

000010111010101000010111101010	1
000100110011101000100110011110	1
000010100011101000010100011110	7
000111001111111000111001111111	8
001111111110101001111111111010	1
000100111001011000100111001101	5
001011011011101001011011011110	1
000001111000011000001111000001	6
1000010000100001000	8
000010001000101000001000100010	9
000100010010011000100010010001	9
001001001111011001001001111101	8
001111010101101001111010101110	1
000110011111110100011001111110	1
000110110100101000110110100010	4
001010111111101001010111111110	1
000011011110101000001101111010	1
000011100110011000001110011001	7
001100110110101001100110110101	1
000001101011101000000110101110	1
000011101100101000001110110010	4
001011101001101001011101001110	1
000001011001101000000101100110	1
000000111101101000000011110110	1
000000100100101000000010010010	8
10010010010010010	7
001001010110011001001010110001	3
001001011100101001001010111001	1
000000010110101000000001011010	10
000101101001001000010110100100	5
001001110111101001001110111110	1
000111001001101000111001001110	8
001001101110101001001101110110	1
000110101101101000110101101110	1
000011001011001000001100101100	2
000101011011001000101011011100	2
000010101111001000010101111100	2
000111110111001000111111011100	2
000110010011001000110011001100	2
000100111111001000100111111100	2
000001100111001000000110011100	2
000000000100111100000000100111	10
000010011101001000001001110100	2
000000000101011000000000010101	10
000001010101001000000101010100	2
000011010010001000001101001000	3
000010110110001000010110110000	3
000001111111000100000111111000	3
000000011010001001000000001101000	3
000000100110000100000100110000	4
000000101000001000000010100000	5
000000000000000	10
000000000011001000000000001100	10
000000110001001000000011000100	6
000100011000101000100011000010	5
000101010001111000101010000111	6
000111111011101000111111101110	1
000101001110101000101001110100	8
001110101110111001110101110111	0
000100101010101000100101010101	9
010101010111011010101011101	0
000100101100111000100101100111	0
000011110011111000011111001111	6
001111010011111001111010011111	6
001111101101011001111110110101	0
000010111100111000010111110011	0
000011000111101000011000111110	8
000111100100111000111100100111	7
001001111101011001001111110101	0
000101101111011000101101111101	0
000010011011011000001001101101	0
000010010111111000010010111111	9
001011111010011001011111101001	0
000010010001101000010010000110	6
000110100111011000011010011101	7
001110110111111100111011011111	0
000010001110111100001000111011	9
001111011111011001111011111101	0
000101011101011000101011101101	0
000011001101011000001100110101	8
001101010100111001101010010011	0
000001110100111000001110100111	0
000101010111101000101010111110	8
010111101110111010111110111011	0
000001101101111100000110110111	0
000101110110011000101110110001	0
000100110101111100010011010111	6
010111110111111010111111011111	0
000010101001011000010101001101	8
1010010100101001010	6
001010110101011001010110101101	0
000001011111111100000101111111	0
000011101010111100001110101011	0
000001100001011000000110000101	7
000010100101111100001011010111	7
001011110110111001010111011011	8
11011011011011011	0
000001010011011000000101001101	6
001101100110111001101100110011	0
000101111100101000010111110010	4
001010111001111001010111001111	8
1110011100111001110	6
001110111011011001110111011101	0
000001001010011000000100101001	8
001010011110011001010011111001	0
000001000110111000000100011011	8
000110110010111000011011001011	6
001011101111111001011101111111	0
000110011001111100011001100111	0
000000111011111000000011101111	0
000000110111011000000011011101	0
000000111001010100000111001010	6
001010100110101001010010011010	7
001101001101111001101001101111	8
001101111111111001101111111111	9
111111111111111	0
000000101110011000000010111001	0
000011010100011000001101010001	4
000101111010111000101111101011	0
000111010110111000111010111011	0
000000100010111000000010001011	7
000101100011111000010110001111	7
000111111101111000111111110111	8
1111011110111101111	0
000000011100011000000001110001	4
000101001000111000101001000011	5
000110101011111000110101011111	7
010111111011011010111111101101	0
000011110101101000001111010110	8
1101011010110101101	0
000101100101101000101100101110	7
001011011101111001010110111011	0
000000010000111000000001000011	6
000011011000111000001101100011	7
1100011000110001100	7
000110010101011000110010101011	8
010101011011111101010101101111	0
000000001111101000000000111110	10
000011111001001000001111100100	5
001001011010111001001011011011	0
000110111110011000110111111001	0
000011111111101100001111111101	0

The total number of bits in the one hundred forty-four sequences is 4064 and the total number of overlaps is 548. This yields a $(15, 1)$-CS of length $4064 - 548 = 3516$.

APPENDIX D
$(16, 1)$-CS FROM SELF-DUAL SEQUENCES OF LENGTH 32

11100100110100110001101100101100111001001101000100011011001011101110010011010 01	11
01001101001000101011001011011101010011010010011010110010110110010100110100100 01	11
11010010001101100010110011001001110100110011011000101101110010011101001000110 11	9
10001101101001010111001001011010100001011010010101111010010110101000110110100 10	9
01101001010000011001111010111110011000010100000110010110101111100110100101000 00	9
01010000001011001010111111110011010100000000011001010111111010011010100000010 110	8
00010110100000101110100101111101000100101000001011101101011111010001011010000 01	7
10000011110001010011110000111010110000111100010101111100001110101000001111000 10	5
00010111100001111110100001111000000100111000011111101100011110000001011110000 11	10
11110000110110000000011100100111111110001101100000001111001001111111000001101100	8
01101100100010001001001101110111011011011000100010010010011101110110110010000100	8
01000100001011111011101110100000100010000101011101110111010100010001000010111	12
00100001011100001100111010001111001100010111000011011110100011110010000101110000	13
10000101110001000110101000111011100101011100010001111010001110111000010111000010	12
00101110001010001101000111010111011011100010100010010001110101110010111000010100	10
11000101001000010011101111011110111010001000010000100111010110111101110001010010000	8
10010000100010100110111101110100100100001000101101101111011101011001000001000101	7
10001011101111000111010001000001100010111011111001110100010000111000101110111110	8
11011110001001110010000111011000110111100110011100100001100110001101111000010011	7
00100110011000001101100110011111101100110011000001001100110011111100100110011000000	10
11001100001101010011001111001010110011000111010100110011100010101100110000011010	9
00001101011111111111100101000000000001101111111111111100100000000000001101011111	10
10101111110000000101000000111011101011111100010001010000001111111010111111000000	10
11111000001100000000010111001111111111010001100000000011111001111111110000011000	10
00000110001100011111100111001100000011000110011111110011100110000000011000110000	11
01100011000010001001110011111001101100011000011001001110011111011101100011000000100	9
11000010011011000011110110010011110000100111110000111101100000111100001001101110	10
01001101101101001011001001001011010011011111010010110010000010110100110110101010	9
01101101010011011001001010110010011001010100110110011010101100100110110101001100110	7
01001100000011011011001111110010010111000000110110100011111100100100110000000110	12
01100000011001101001111110011001001000000110011011011111100110010110000000110011	9
00011001100110101110011001100101001110011001101011000110011001001010001100110011101	6
00110100011100101100101110001101011101000111001010001011100011010011010001110011	11
01000111001110111011100011000100010001110001101110111000111001000100011100111101	10
11100111010111100001100011100001111001110001111000011000101000011110011101011111	9
11010111101010110010100001110100110101111000101100101010000101010011010111101010 1	7
10101010111111011010101010000010010111010111110110100010100000100101010101111101	10
01011111010100101010000010101101111011111010100100010000010101101010111110101001	8
10101001001110010101001011000110101011010011100101010110110001101010100100111100	10
00100111000100101101100011101111001001110001000011011000111011010010011100011001	11
01110001001000001001111011011111101100001001000001000111011011111011100010010000	11
00010010000101001010110111101011010100100000101001110110111101011000100100001010	9
10000101011000000011110111001111110000100011000000111101010011111100001010110000	11
01010110000010101010100111010101010101100010101010101001111101010101011000001010	10
11000001010000100001111010111101111000010100001000111110101111011100000010100001	5
00001010010110111111101011010010000011010010110111111001011010010000001010010110 1	13
00101001011010001101011010010111001110010110100011000110100101110010100101101000	8
10110100001001000100111111011011011000000010010001001011110110111011010000010010	13
11010000100101100110111101101001100100001001011000101111011010011101000001001011	5
01011011110011111010010000110000010111111100111110100000001100000101101111001111	10
01111001111011011000001000010010011111011110110110000110000100100111100111101110	9
01111011001010101000010011010101011010110010101010010100110101010111101100101010 1	13

1110110010101101000100110101001011111100101011010000001101010010111011001010110	10
1001010110001111011010100111000110010101100011100110101001110000100101011000111	7
1000111101111101011100001000001010001111111111010111000000000010100011110111110	9
1101111100010111001000001110100011011101000101110010001011101000110111110001011	12
1111100010111110000001110100000110111000101111100100011101000001111110001011111	11
1000101111110111011010000001000100010111101011011011010000101000100010111111011	8
1111101101101111000001001001000011111111011011110000000010010000111110110110111	10
0110110111010011100100100011110001101101110000111001001000101100011011011101001	9
0111010010101011100010111010111000111010010100011100010110101010001110100101010 1	8
0101010110011111101010100110000001010111100111111010100001100000010101011001111	12
1010110011110000010100110000111110101100101000001010011001011111010110011110000	11
1100111100010000001100001110011111001111000110000011000011101111110011110001000	0

The total number of bits in the sixty-four sequences is 5056 and the total number of overlaps is 594. This yields a $(16, 1)$-CS of length $5056 - 594 = 4462$.

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